ISYS 1108 / 3474 Software Engineering Project Management

Week 3: Workshop

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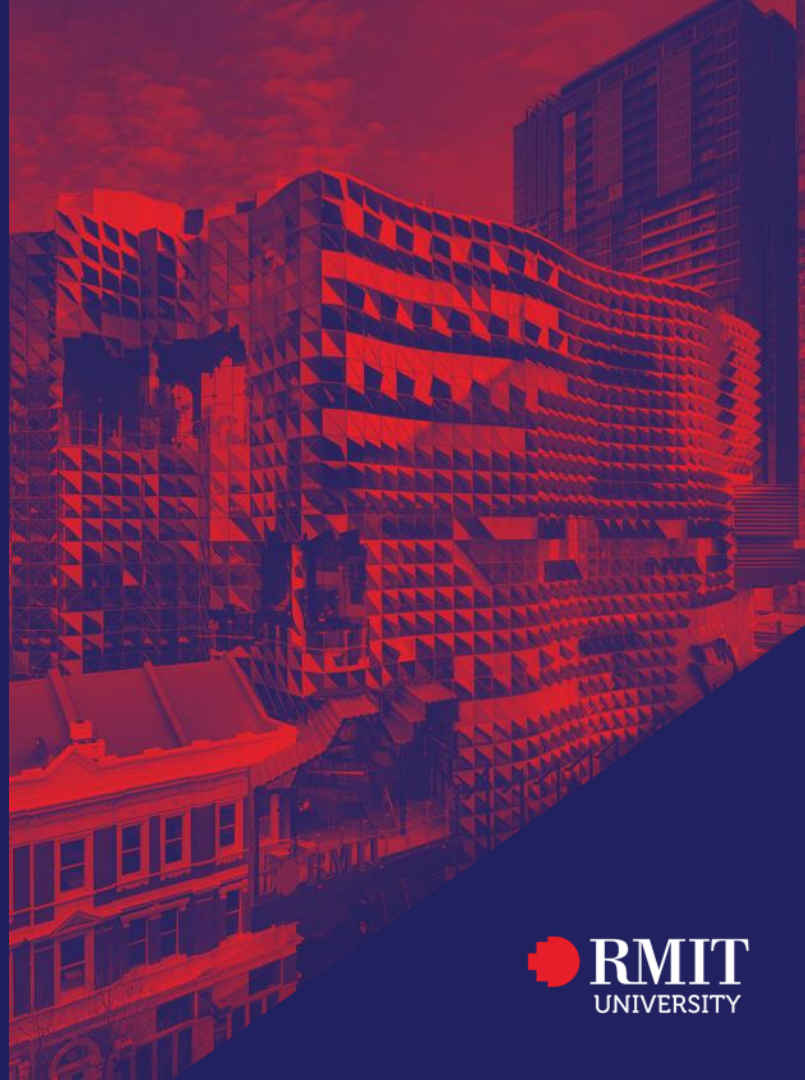
What's next...



RMIT UNIVERSITY

Acknowledgement of Country

RMIT University acknowledges the people of the Woi wurrung and Boon wurrung language groups of the eastern Kulin Nation on whose unceded lands we conduct the business of the University.

RMIT University respectfully acknowledges their Ancestors and Elders, past and present.

RMIT also acknowledges the Traditional Custodians and their Ancestors of the lands and waters across Australia where we conduct our business.

Our plan for this session

- 😊 Recap: V&S and PC
- 😊 Victoria Farms Scenario – drafts of Product backlog parts, creating a full product backlog
- 😊 Sample for SE Challenge 1

Please use this link to join EdForum using your RMIT student email address

https://edstem.org/au/join/6tzstn

If you are an exchange student or have RMIT Vietnam student email address, please send an email to Dr Maria Spichkova (maria.spichkova@rmit.edu.au) to be added to the EdForum manually.

Recap: We always use only one of these documents!

Vision & Scope document:

1. **Business Requirements**
   1) Background
   2) Business Opportunity
   3) Business Objectives and Success Criteria
   4) Business Assumptions and Dependencies
   5) Business Risks
2. **Vision of the Solution**
   1) Vision Statement
   2) Major Features
3. **Scope and Limitations**
   1) Scope of Initial and Subsequent Releases
   2) Limitations and Exclusions
4. **Business Context**
   1) Stakeholder Profiles
   2) Project Priorities
   3) Operating Environment

Project Charter (PMBOK)

1. Purpose *
2. Objectives *
3. Overview *
4. Schedules *
5. Resource requirements
6. Personnel and stakeholders *
7. Evaluation methods
8. Risk management *

Activity 1: Victoria Farms scenario

Duration: approx. 10 minutes

This week focus on the following case study: Victoria Farms.
Read, analyse and discuss the scenario: general description and functionality to implement within the project.

A team created very early drafts of the *Product Backlog* for the above product, but they are incomplete and contains mistakes – our next activity will focus on analysing and fixing these drafts. For this analysis and correction we need to know the scenario.

Victoria Farms scenario: General description

“Victoria Farms” is an innovative technologically driven Food Service distributor who uses the strategy of franchising to make a significant difference in distributing Fruit, Vegetables and Dairy into the Hospitality Industry. “Franchisees” are drivers who pick up the fresh produce from the Victoria Farms warehouse, pack the produce, and deliver it to the restaurants that are customers of Victoria Farms. They work 7 days a week.

Victoria Farms’ technology has been specially developed for their business and is largely a web ordering, procurement, and distribution system.

There has been interest from parties in similar businesses and industries who would like to adopt the technology and so Victoria Farms would like to begin to make the software a commercial product, by initially doing a full rewrite of the existing interfaces.

Victoria Farms scenario: Required functionality

The major system functions required are:

- Victoria Farm's user maintenance system for maintaining records of all the system's users
- Inventory system for all available fresh produce.
- Maintenance of all franchisee and customer records, these include
  - Types of orders
  - Dates of orders
  - Details of order deliveries
  - Details of product returns
  - Cost of products purchased
  - Payment of products purchased
  - Monthly financial reports

Activity 2: Draft 1 of Product backlog

Duration: approx. 10 minutes

Let’s now analyse the first draft of a backlog’s part. You can also use the checklist for feedback on Product backlog as a guidance for your analysis.

Is our Product Backlog correct?

- ✓ Are the PBI IDs unique? Do the IDs make sense?
- ✓ Are the PBI written as reasonable user stories? (reasonable semantics, correct format As a… I want to … so that …)
- ✓ Is the product backlog sorted by the priorities of the PBIs?
- ✓ Are the priorities correct wrt. notes made during the initial requirements elicitation interviews with the stakeholders?
- ✓ Are the Owners reasonable wrt. the PBIs?
- ✓ Are the Efforts reasonable wrt. the PBIs?
- ✓ Are the DoDs reasonable wrt. the PBIs?
- ✓ **Is the PB complete?**

Draft 1 of Product Backlog: What exactly should be fixed and why?

ID	ITEM	OWNER	PRIORITY	EFFORT in story points	STATUS	DEFINITION OF DONE
8	Users: Ability to updated customer payment method.	Anna	High	21	To Do	Unit testing
5	Users: Ability to add franchisee	Sam	Medium	3	To Do	Functional testing based on Acceptance criteria (80% coverage)
6	Users: Ability to maintain franchisee accounts (Contact Details)	John	Medium	1	To Do	Integration testing
7	Users : Ability to update types of orders for customers.	Lars	Low	21	To Do	Users can accept or decline shifts allocated to them by a manager. If declining, a note with explanations should be provided.

- ✓ Are the PBI IDs correct?
- ✓ Are the PBI written as reasonable user stories?
- ✓ Is the product backlog sorted by the priorities?
- ✓ Are the priorities correct?
- ✓ Are the Owners reasonable wrt. the PBIs?
- ✓ Are the Efforts reasonable wrt. the PBIs?
- ✓ Are the DoDs reasonable wrt. the PBIs?
- ✓ **Is the PB complete?**

Activity 3: Draft 2 of Product backlog

Duration: approx. 10 minutes

Let’s now analyse the second draft of a backlog’s part. You can also use the checklist for feedback on Product backlog as a guidance for your analysis.

Draft 2 of Product Backlog: What exactly should be fixed and why?

ID	ITEM	OWNER	PRIORITY	EFFORT in story points	STATUS	DEFINITION OF DONE
5	Users: Ability to add franchisee	Product Owner	Medium	8.00	To Do	Victoria Farms Admin user can add Franchisee to the system.
6	Users: Ability to maintain franchisee accounts (Contact Details)	Product Owner	Medium	2.00	To Do	Victoria Farms can updated Franchisee Contact details. Franchisee can update their contact details.
8	Users: Ability to updated customer payment method.	Product Owner	High	4.00	To Do	Victoria Farms can updated Franchisee Payment Method via Admin Console.
7	Users : Ability to update types of orders for customers.	Product Owner	Low	2.00	To Do	Victoria Farms can updated Franchisee Order Type via Admin Console.

- ✓ Are the PBI IDs correct?
- ✓ Are the PBI written as reasonable user stories?
- ✓ Is the product backlog sorted by the priorities?
- ✓ Are the priorities correct?
- ✓ Are the Owners reasonable wrt. the PBIs?
- ✓ Are the Efforts reasonable wrt. the PBIs?
- ✓ Are the DoDs reasonable wrt. the PBIs?
- ✓ **Is the PB complete?**

Activity 4: Fixing this part of the Product backlog

Duration: approx. 10 minutes

Let’s now fix this part of the product backlog. You can also use the checklist for feedback on Product backlog as a guidance for your analysis.

Templates for a Product backlog can be downloaded from
Modules → Templates, Tools, Team Allocation → Useful templates

Sample Correction for this part of Product Backlog

PRODUCT BACKLOG

IMPORTANT:
We fixed here all points except completeness issue.
How would you fix completeness issue?

DoD1:
- Unit tests passed (code coverage 85%)
- Code peer reviewed
- Functional test based on ACs passed
- Regression testing completed
- Usability testing completed on Google Chrome (version 89) and Safari (version 14)
- Approved by the Product Owner

ID	ITEM	OWNER	PRIORITY	EFFORT in story points	STATUS	DEFINITION OF DONE
8	As a customer user, I want to update my payment method, so that my payment details are up-to-date.	Product Owner	High	2	To Do	DoD1
5	As a new franchisee user, I want to create a franchisee account, so that I can start working with Victoria Farms	Product Owner	Medium	8	To Do	DoD1
6	As a franchisee user, I want to update the contact details in my account, so that my customers can contact me	Product Owner	Medium	2	To Do	DoD1
7	As a customer user, I want to update the type of my order, so that the deliverer knows whether the order is urgent.	Product Owner	Low	2	To Do	DoD1

Completeness of the Product Backlog (PB)

We cannot measure completeness in terms like "how many PBIs are in the backlog"

- **It's not about the number of PBIs in PB**
- **It's about coverage of all the requirements**

In the case of assignment project:

- You have to cover all requests listed in the *notes made during the initial requirements elicitation interviews with the stakeholders.*
- It might be complicated to cover all this information in less than 20 PBIs…

Completeness of the Product Backlog, for example…

- 30 PBIs, all requirements covered, no out of scope requirements
- 60 PBIs, all requirements covered, no out of scope requirements

- 30 PBIs, all requirements covered, some requirements are out of scope
- 30 PBIs, not all requirements covered, no out of scope requirements
- 60 PBIs, all requirements covered, some requirements are out of scope
- 60 PBIs, not all requirements covered, no out of scope requirements
- 30 PBIs, not all requirements covered, some requirements are out of scope
- 60 PBIs, not all requirements covered, some requirements are out of scope

Product backlog is complete ==
it covers the expected scope of the project

Activity 5: Creating a complete Product backlog

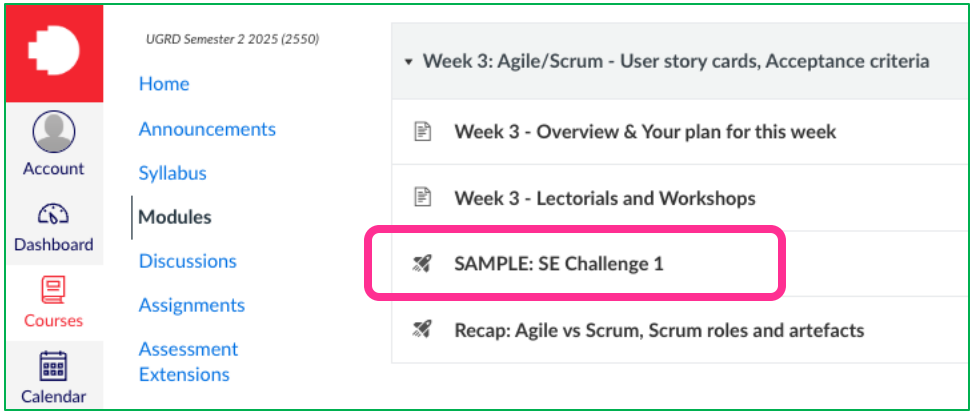
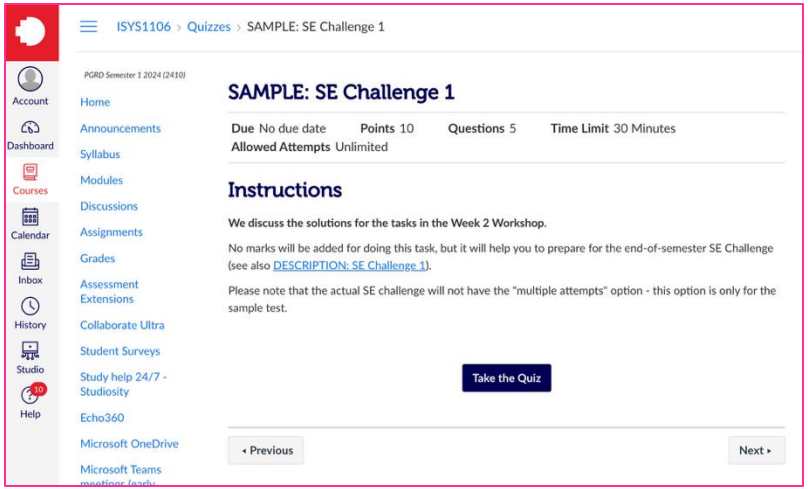
Duration: approx. 40 minutes

Let's now try to create a complete product backlog for the Victoria Farms scenario.
Note that there will be no sample solution for this activity, but the tutor can provide you feedback on your solution within this session.

Templates for a Product backlog can be downloaded from
Modules → Templates, Tools, Team Allocation → Useful templates

Activity 6: Sample Challenge 1

Duration: approx. 30 minutes

Try to complete sample challenge. You will see the correct solutions as soon as you complete the challenge.

Ask your tutor if you have any questions.